Ethical Considerations in Automated Agricultural and Environmental Control Systems

Computer Control Systems — Spring 2025/2026

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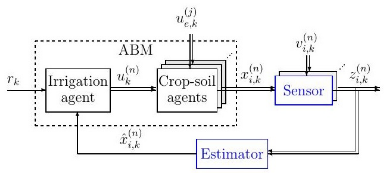

Figure 5.1: Blocks diagram of a second version of the ABM with a reduced number of sensors and a single estate estimator.

Figure 1: ABM with limited sensors and single state estimator. Sensor output z is the sole field-to-estimator link [1].

I. FEEDBACK, SENSING, AND ETHICAL RISK

A. *Sensor Placement and Spatial Ethics*

Choosing where to place sensors across a 20,000-donum zone is ultimately a decision about *who bears risk*. Lopéz's agent-based EKF framework [1] captures this tension in a single cost-performance objective:

$$J(\mathcal{S}) = \mathrm{tr}(P_k(\mathcal{S})) + \lambda\,|\mathcal{S}|\,c_s \tag{1}$$

Raising λ cuts sensor count and cost, but grows uncertainty in the drainage flux $\varphi_3^{(n)}$ — the primary leaching pathway for pesticides and nitrogen toward the shared aquifer. This also introduces *unequal monitoring*: economically important agricultural zones may receive denser sensing coverage while environmentally sensitive regions near community groundwater sources remain insufficiently observed. When cost savings accrue to the operator while contamination risk falls on neighboring communities, a **soft trade-off becomes a non-negotiable ethical violation**. This demands a hard constraint alongside (1):

$$\forall\, n \in \mathcal{S}_{\mathrm{crit}} : \delta_{n,k} = 1 \tag{2}$$

Patches at drainage boundaries and aquifer recharge zones belong to $\mathcal{S}_{\mathrm{crit}}$ and must remain sensed regardless of λ. Community co-design in selecting these patches is not optional goodwill — it is a *structural accountability requirement* [2]. In practice, this means establishing a joint technical committee that includes both agricultural engineers and community water representatives, with formal veto authority over any sensor placement plan that leaves recharge zones unmonitored. Without this mechanism, the constraint in (2) exists only on paper.

B. *Sensor Bias, Model Mismatch, Climate Change, and Silent Harm*

The drainage coefficient β in the water-balance model

$$x_{1,k+1}^{(n)} = x_{1,k}^{(n)} - \varphi_{1,k}^{(n)} - \varphi_{2,k}^{(n)} - \varphi_{3,k}^{(n)} + \varphi_{4,k}^{(n)} + u_{e,k}^{(1)} + u_{c,k}^{(n)} \tag{3}$$

accounts for 36% of total output variance — the single most influential parameter [1]. Under historical climate it is identifiable within $\pm 11\%$; under climate change, shifting rainfall patterns and altered soil hydraulics invalidate that calibration. The control system then issues irrigation commands without error flags, yet systematically underestimates leaching — what we call **silent model-induced harm**. USGS data show that pesticide-contaminated groundwater can take *decades* to recover after inputs are reduced [3], making model mismatch a fundamentally different problem from a reversible engineering error.

Under IPCC projections for the Jordan Valley and similar semi-arid regions, mean annual precipitation may decrease by 10–20% by 2050 while extreme rainfall events intensify [4] — precisely the conditions that invalidate a β calibrated under historical data. A practical safeguard is a *drift threshold*: if re-identified β deviates by more than $\pm 20\%$ from its calibration value, the system must automatically flag the irrigation schedule for human review before execution rather than applying it autonomously.

The Wisconsin neonicotinoid case [6] illustrates this at scale: between 2013 and 2015, 78% of 91 high-capacity irrigation wells tested positive for thiamethoxam (mean $0.28\,\mu$g/L), not because any system malfunctioned, but because no monitoring existed at drainage boundaries. Contamination had already propagated through the aquifer before detection — the exact outcome of deploying (3) without constraint (2).

Remote sensing inputs add a second risk layer. When optical sensors degrade under clouds — precisely when monitoring matters most — a static measurement covariance $R_k^{(n)}$ causes the EKF to over-trust corrupted data: **confident, wrong, and invisible** [5, 7]. The sensor equation is

$$z_k^{(n)} = C^{(n)} x_k^{(n)} + H^{(n)} u_{e,k}^{(n)} + v_k^{(n)}, \quad v_k^{(n)} \sim \mathcal{N}\Big(0, R_k^{(n)}\Big) \tag{4}$$

If $R_k^{(n)}$ is not updated to reflect degraded sensor quality, the filter over-weights corrupted observations and produces biased state estimates that drive incorrect control actions.

C. *Observer, Fault Detection, and Ethical Harm Reduction*

The EKF innovation

$$\tilde{e}_k = z_k - C_k \hat{x}_k^- \tag{5}$$

acts as the system's built-in *ethical sensor*: sustained non-zero mean or growing variance flags sensor drift or model mismatch [7]. A formal decision rule uses the *Normalized Innovation Squared* (NIS):

$$\mathrm{NIS}_k = \tilde{e}_k^\top S_k^{-1} \tilde{e}_k \tag{6}$$

If NIS_k exceeds the $\chi^2(n_z,\, 0.95)$ threshold for three or more consecutive time steps, the system must halt autonomous irrigation commands and alert a human operator — converting a statistical signal into an auditable accountability checkpoint. Four engineering practices translate this signal into active harm prevention:

1. **Adaptive R-tuning.** Inflate R_k during RS outages. Jihani et al. [5] demonstrated 99.9% fault-detection precision in irrigation WSNs using KF combined with an autoregressive sensor model — achievable only with adaptive, not static, covariance settings.

2. **Q-inflation under climate anomalies.** Lopéz decomposes $Q = Q_p + Q_g$ [1], where Q_g encodes inter-patch flow uncertainty via graph adjacency. When climate statistics deviate from calibration baselines, inflating Q_p prevents overconfident drainage estimates that would mask aquifer recharge risk.

3. **Pre-deployment observability gate.** The Jacobian $\mathcal{O}(x,u) = \partial q/\partial x$ must satisfy $\text{rank}(\mathcal{O}) = n_x$ before system authorization [1]. If drainage flux φ_3 is unobservable under the chosen sensor set, deployment near shared water resources is ethically impermissible — regardless of cost or schedule.

4. **Chemical software sensors.** The nitrogen balance
$$\frac{dN_s}{dt} = N_m + \alpha N_R - N_{\text{uptake}} - N_l \quad (7)$$
must be embedded as an observer state [1]. Shekhar et al. [8] showed that biased NPK estimation from static filter tuning produced excessive fertilizer recommendations and environmental contamination with no system alarm. Without a chemical observer, the ethical blind spot is structural.

Table 1 summarises the five principal risk pathways, their failure modes, ethical consequences, and whether mitigations are negotiable soft constraints or non-negotiable hard requirements.

Table 1: Ethical Risk Mapping: Sensor and Model Failures

Risk Factor	**Failure Mode**	**Ethical Consequence**	**Mitigation (Soft / Non-Neg.)**
Sparse placement (λ high)	φ_3 unmonitored near aquifer	Community water contamination	**NON-NEG.:** $\mathcal{S}_{\text{crit}}$ constraint (2)
Static R_k under RS outage	Corrupted estimates accepted	Over-irrigation $\rightarrow$ leaching spike	Soft: adaptive R-tuning
β mismatch under climate shift	Drainage underestimated	Pesticide leaching; decades of harm	**NON-NEG.:** Q-inflation + re-ID β
No chemical observer	N/pesticide flux unobserved	Silent progressive contamination	Soft: embed N_l software sensor
$\text{rank}(\mathcal{O}) < n_x$ at deployment	φ_3 unobservable	Hazard invisible to system	**NON-NEG.:** block deployment

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